Topic 18 — World Landforms

Covers syllabus (click to expand)

Continents (area · countries · elevation) | Mountains (fold / block / volcanic / residual) | Plateaus | Valleys | Plains | Volcanoes | Islands | Deserts | Grasslands | Vegetation | World Climatic Regions

Sources baked in: Lucent world-geography match-lists; NCERT; Drishti; **Ghatnachakra** World Geography Continents–Mountains–Plateaus–Valleys–Deserts–Grasslands–Islands (**CB–60+** / **CB–63+** / **CB–75+** / **CB–81+** / CB–82+ / CB–145+); UPPCS Prelims 2018–2025 **Weight:** ★★★ — continent ranks, mountain–country, desert size/country, grassland names, Med winter rain, volcano location **Last verified:** August 2026 — GC continents + landform pass **Current Affairs:** none mandatory (static map chapter)

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Static	Andes longest vs Himalaya highest	Fold-chain swap	Atlas
Static	Kilimanjaro = East African Rift (not Ring of Fire)	Volcano belt trap	Coaching
Static	Hot-desert west-coast + cold-current logic	Desert location stems	NCERT
Static	Grassland name set (Pampas / Campos / Llanos / Prairie / Steppe)	Match-list diet	Lucent

N.0 Framework — World Landforms & Natural Regions

Identity: This chapter is Lucent **map-matching** for landform → country, desert → size/country, grassland → local name, and climate → rain timing.

Landforms set the physical skeleton.

Mountains, plateaus, plains, volcanoes, islands and deserts come first.

Biomes sit on that skeleton.

Grasslands, vegetation belts and climatic regions come after the landform cards.

Do not mix a soil name with a forest name.

Terra Roxa is not the same fact as **Selva**.

Mountain questions need **family** plus **country**.

Family means fold, block, volcanic or residual.

Desert questions need **hot versus cold**, the **size ladder**, and the **west-coast + cold-current** why-card.

Order of this chapter

1. Continents → mountains → plateaus → valleys → plains.
2. Volcanoes → islands → deserts.
3. Grasslands → vegetation → world climatic regions.

Learn the full mountain, desert and grassland match set below. Any standard range or pair can appear, not only Kilimanjaro or Gobi.

N.0A Continents

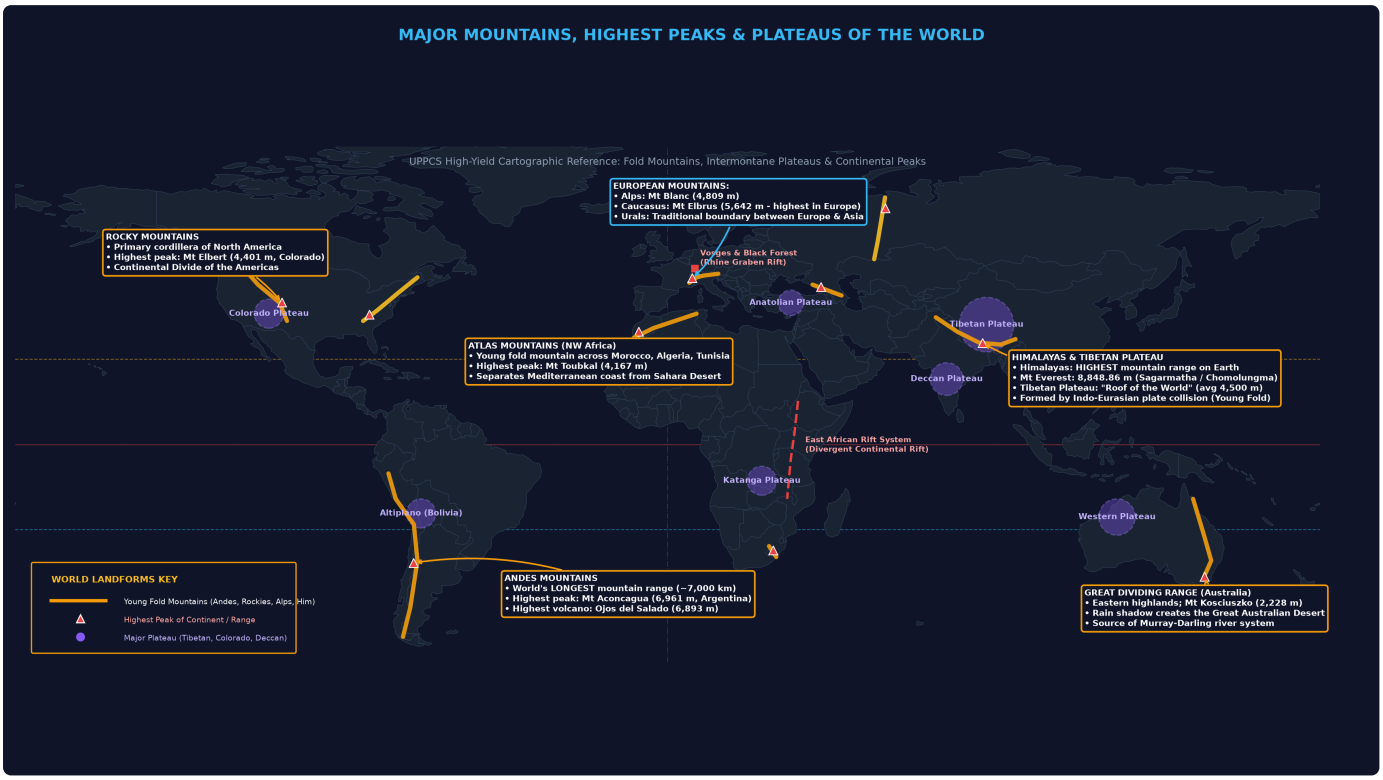
Seven continents by **area** (largest → smallest):

Rank	Continent	Area pair (km²)	% of Earth's land (Philip's-type)	Countries (GC sheet)
1	**Asia**	~44.6 million	~29.8%	~48
2	**Africa**	~30.1 million	~20.3%	~**54** (most of any continent)
3	**North America**	~24.5 million	~16.2%	~23
4	**South America**	~17.8 million	~11.8%	~12 (Brazil largest)
5	**Antarctica**	~14.2 million	~9.2%	—
6	**Europe**	~9.9 million	~6.8%	~49
7	**Australia**	~7.7 million	~5.9%	Oceania ~14 in wider lists

- The **smallest continent** is Australia.
- The **two smallest** among classic sets are **Australia and Europe**.
- **Highest mean elevation** belongs to **Antarctica** (about **2300 m**). Approximate coaching means after that are Asia (~923 m), North America (~760 m), Africa (~585 m), South America (~554 m), Australia (~330 m), and Europe (~302 m).
- The **highest share of plains** in total area belongs to **Europe** (then South America among large shares).
- **Maximum per capita land** in World Bank-type coaching belongs to **Australia**.
- Africa alone is crossed by **Cancer, Capricorn and Equator**.
- **Guyana** is in South America (Georgetown; English is official). Do not place it in Africa with Gabon, Guinea or Guinea-Bissau.
- Asia's borders run along the Urals–Caspian–Black Sea–Mediterranean versus Europe, the Red Sea / Suez versus Africa, and the Bering Strait versus North America.
- The Panama Canal separates North and South America.
- Latin America framing covers South America, Middle America, Mexico and the West Indies.
- Antarctica is the coldest, driest and windiest continent, with the highest average elevation.

Order for this chapter after continents: mountains → plateaus → valleys → plains → volcanoes → islands → deserts → biomes.

N.1 Mountains



Four (चातुर्याम) families

Family	Force / story	World examples
Fold	Compression (orogeny)	Himalaya, Alps, Andes, Rockies, Atlas, Pyrenees, Zagros. The **Appalachians** are an old worn fold
Block / fault / rift	Tension / uplifted blocks	Vosges, Black Forest, Sierra Nevada; African rift shoulders (**Kilimanjaro, Kenya, Rwenzori**)
Volcanic cone	Magma	Fuji, Etna, Cotopaxi, Mauna Loa — location table in **N.4**
Residual / dome	Old highland left after long denudation; or magma push (dome)	Residual leftovers of old folds (Appalachian-type worn highlands); **Black Hills** / laccolith **dome**

Highest peak by continent (match-list gold)

Continent	Peak	Range	Country fact
Asia	**Everest** (8848 m)	Himalaya	Nepal–China
Asia (2nd)	**K2 / Godwin Austen**	Karakoram (काराकोरम)	Pakistan–China

Continent	Peak	Range	Country fact
South America	**Aconcagua**	Andes	**Argentina** — highest outside Asia
North America	**Denali / McKinley**	Alaska Range	USA
Africa	**Kilimanjaro**	—	**Tanzania** (not Kenya)
Europe	**Elbrus**	Caucasus	Russia
Alps (W Europe)	**Mont Blanc**	Alps	France–Italy
Australia (mainland)	**Kosciuszko**	Snowy Mts	Australia
Oceania (often asked)	**Puncak Jaya / Carstensz**	Sudirman	Indonesia (New Guinea)
Antarctica	**Vinson Massif** (4897 m)	Ellsworth Mts	Deepest point: Bentley Subglacial Trench (-2555 m)

Continental Extremes: Highest and Lowest Points

Continent	Highest Point (Peak & Elevation)	Lowest Point (Depression & Elevation)
Asia	**Mount Everest** (8,848 m)	**Dead Sea** (-397 m / -400 m below sea level)
Africa	**Mount Kilimanjaro** (5,895 m)	**Lake Assal** (-156 m, Djibouti)
North America	**Denali / Mt. McKinley** (6,194 m)	**Death Valley** (-86 m, California)
South America	**Mount Aconcagua** (6,960 m)	**Valdes Peninsula** (-40 m, Argentina)
Europe	**Mount Elbrus** (5,642 m)	**Caspian Sea shore** (-28 m)
Australia	**Mount Kosciuszko** (2,228 m) / Puncak Jaya (4,884 m, Oceania)	**Lake Eyre** (-16 m)
Antarctica	**Vinson Massif** (4,897 m)	**Bentley Subglacial Trench** (-2,555 m)

Major ranges — country / trap

Range	Where	Highest / note
Andes	W South America, ~7,000 km	**Longest** fold chain. Chimborazo (**Ecuador**). Copper Chile–Peru. Rain-shadow → Atacama + Patagonia
Himalaya	India–Nepal–Bhutan–China	**Highest**. Young Tertiary. Everest / Kanchenjunga (कंचनजंगा)
Karakoram (काराकोरम)	PK–India–China	**K2**. Not a Himalaya “range name” swap
Rockies	W Canada–USA–Mexico fringe	Young cordillera. **Not** the Appalachians

Range	Where	Highest / note
Appalachians	**Eastern** USA	**Old** fold; worn; coal fields
Alps	France–Switzerland–Austria–Italy	**Mont Blanc**. Switzerland is the classic Alps country. The foehn is a warm downslope wind on the **north** side
Pyrenees	**Spain–France** (Andorra)	**Not** Alps
Apennines	**Italy**	Spine of the peninsula
Balkans	**Bulgaria** / SE Europe	Not Carpathians
Carpathians	Romania / Slovakia / Ukraine	Transylvania arc — do not place on Bulgaria
Caucasus	Russia–Georgia–Azerbaijan	**Elbrus** is Europe's high point
Urals	Russia	Conventional **Europe–Asia** divide
Atlas	Morocco–Algeria–Tunisia	Highest peak **Toubkal** is in **Morocco** (not Algeria)
Drakensberg	South Africa / Lesotho	SE African escarpment
Ethiopian Highlands	Ethiopia	Blue Nile / Lake Tana source side
Rwenzori	Uganda–DRC	**Mt Stanley** — country fact **Uganda**
Hoggar / Ahaggar	**Algeria**	Sahara highland. **Not** Atlas
Tibesti	Chad	Sahara highland
Zagros / Elburz	Iran	Zagros is the west fold. Elburz is the north range, with **Damavand**
Tien Shan / Altai	C Asia	Syr Darya / Ob headwaters
Kunlun	N of Tibet	Tibet's northern wall
Great Dividing Range	Eastern Australia	Old; not Himalaya-young; ~3700 km; Mt Kosciuszko
Southern Alps	South Island, **New Zealand**	Aoraki / **Mt Cook**
Darling Range	**SW coast** of Australia (Western Australia)	Not NE Australia
Arakan Yoma	**Myanmar**	India–Myanmar hills: Naga, Chin, Patkai / Lushai; peak Natma Taung / Mt Victoria — **Khasi is Meghalaya (मेघालय)**, not this border trio
White Mountains	New Hampshire / Maine, **USA**	
Black Mountains	North Carolina, **USA** (Appalachian)	Not Black Forest

Range	Where	Highest / note
Black Forest	**Germany** (Feldberg); east of **Rhine**; Vosges west in France	Block mountain
Salt Range	Pakistan	Block mountain note
Allegheny	USA (Appalachian)	
Cantabrian	**Spain**	
Mackenzie (मैकेंज़ी)	**Canada**	
Jura	France–Switzerland	
Dinaric Alps	Balkans coast belt	
Pennines	Europe (UK)	Residual / old with Appalachians & Aravallis

Length pair (surface ranges): Andes (~7000–8900 km) > Rockies (~4800) > Great Dividing (~3700) > Himalaya (~2500).

Orogeny age pairs - The Alps, Andes, Himalaya, Rockies and Atlas are **Tertiary / young fold** mountains. - The Appalachians are **Caledonian / old**. - The Aravalli (अरावली) is Pre-Cambrian. - Black Forest, Vosges and Tien Shan-type ranges carry a Hercynian note. - The **Sierra Nevada** is framed as the world's most extensive **block** range (California). It is **not** a young folded mountain like the Rockies, Alps or Himalaya.

Phrase: British Columbia is called the "**Sea of Mountains**" (Molyneux St John / Dufferin (डफ़रिन) tour note).

The **Golan Heights** is a Middle East basaltic plateau in south-west Syria (with a 1967 occupation note). It is not in Central Asia, South-East Asia or Central Africa.

African / Andes peaks that come as a four-row match

Peak	Country	Do not confuse with
Kilimanjaro	**Tanzania**	Kenya (Mt Kenya is a different peak); **not** Ring of Fire
Toubkal	**Morocco**	Hoggar
Stanley	**Uganda**	Kilimanjaro
Hoggar	**Algeria**	Atlas / Toubkal
Chimborazo	**Ecuador**	Peru/Chile Andes dump
Aconcagua	**Argentina**	Chile
Cotopaxi	Ecuador	Volcanic Andes

> **Logic:** ** Paper may print **Kalimanjaro** and **Chimbrozo**. Same peaks. Atlas is **Morocco**, not

> France/Spain (that is Pyrenees).

Inline PYQ — UPPCS Prelims 2024, Q61

Match List-I with List-II and choose the correct answer using the codes given below the lists:

List-I (Country)	List-II (Mountain)
A. Spain–France	1. Alps
B. Switzerland	2. Apennine
C. Bulgaria	3. Pyrenees
D. Italy	4. Balkan

Options:

A. 3 1 2 4

B. 3 1 4 2

C. 2 3 1 4

D. 4 1 2 3

Show answer

Ans: B — A-3 Pyrenees · B-1 Alps · C-4 Balkan · D-2 Apennine. Trap **A** swaps Apennine and Balkan (Italy would get Balkan). Trap **D** gives Spain–France the Balkans.

Inline PYQ — UPPCS Prelims 2020, Q86

Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Mountain)	List-II (Country)
A. Atlas	1. France/Spain
B. Kalimanjaro	2. Ecuador
C. Chimbrozo	3. Tanzania
D. Pyrenees	4. Morocco

Options:

A. 4 3 2 1

B. 4 1 2 3

C. 2 3 4 1

D. 1 3 4 2

Show answer

Ans: A — Atlas–**Morocco** (4) · Kilimanjaro–**Tanzania** (3) · Chimborazo–**Ecuador** (2) · Pyrenees–**France/Spain** (1). Paper spellings **Kalimanjaro / Chimbrozo** are the same peaks. Trap **B** gives Kilimanjaro to France/Spain (that is Pyrenees). Trap **D** gives Atlas to France/Spain.

Inline PYQ — UPPCS Prelims 2023, Q65

Match List-I with List-II and select the correct answer using the code given below:

List-I (Mountain)	List-II (Country)
A. Kilimanjaro	1. Morocco
B. Toubkal	2. Algeria
C. Stanley	3. Tanzania
D. Hoggar	4. Uganda

Options:

A. A-4, B-3, C-2, D-1

B. A-3, B-1, C-4, D-2

C. A-3, B-4, C-1, D-2

D. A-4, B-2, C-3, D-1

Show answer

Ans: B — Kilimanjaro–Tanzania · Toubkal–Morocco · Stanley–Uganda · Hoggar–Algeria. Do not park Toubkal in Algeria (that is Hoggar) or Kilimanjaro in Uganda (that is Stanley).

N.2 Plateaus

A plateau is an **elevated tableland** — flatter than a mountain range, higher than a plain. Rivers often **dissect** it into mesas and gorges.

Kind	Meaning	World example
Intermontane	High block **between** fold ranges	**Tibet** (Himalaya–Kunlun); **Altiplano** (Andes)
Piedmont	At the **foot** of mountains	Appalachian piedmont; Patagonian edge
Continental / shield	Old worn-down block	**Deccan** (दक्कन), Brazilian, African, Western Australian
Lava / volcanic	Flood basalt	**Columbia** (USA), **Deccan Trap** (दक्कन ट्रैप) cover

Kind	Meaning	World example
Plateau	Where	Note
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Tibetan Plateau	China / Tibet	**Highest large** plateau. Headwaters: Indus (सिंधु), Sutlej (सतलुज), Brahmaputra (ब्रह्मपुत्र), Mekong, Yangtze, Hwang Ho
Pamir	Tajikistan knot	Phrase **"Roof of the world"**. Knot of Himalaya–Karakoram–Hindu Kush–Tien Shan
Deccan	Peninsular India	Old gneiss + **Deccan Trap** lava in the west. **Not** Brazil
Malwa / Chotanagpur (छोटानागपुर)	India	Chotanagpur = Jharkhand coal–iron table
Brazilian	Brazil	Coffee on **Terra Roxa**; iron. Separate from Andes
Guiana	Venezuela–Guyana–N Brazil	
Patagonian	S Argentina	Rain-shadow plateau east of Andes
East African / Ethiopian	Africa	Rift lakes; Blue Nile from Ethiopia
African (south)	SA / Botswana / Namibia	High veld sits on it
Colorado	SW USA	**Grand Canyon** (Colorado River) cut into it
Columbia	Washington–Oregon–Idaho	**Lava** plateau
Colorado Plateau ≠ Columbia	—	Canyon vs lava — do not swap
Mexican (Mesa Central)	Mexico	
Anatolian	Türkiye	Interior of Asia Minor
Iranian	Iran	Dasht-e-Kavir / Lut on it
Mongolian	Mongolia	**Gobi** on the dry side
Altiplano	**Bolivia–Peru**	Highest inhabited Andean plateau; Lake Titicaca
Meseta	**Spain**	Interior tableland of Iberia
Loess Plateau	N China	Wind-silt; Hwang Ho yellow colour
Kimberley / Western Plateau	Australia	Old shield
Ozark	USA	Domed plateau note
Shan	Myanmar	Fluvial / tableland note

Kind	Meaning	World example
Telegraphic Plateau	**North Atlantic Ridge** (Newfoundland–Ireland note)	**Not** Indian Ocean

> **Logic:** ** Tibet ≈ **4500 m+** average — closest option often **5 km**. The **Pamir** is stock "Roof of the world" (also applied loosely to Tibet). The **Patagonia Plateau** is "storehouse of minerals" in South America coaching keys. **Madrid** sits on the **Meseta**. Among Coconino / Aquarius / Colorado / Columbia size options, **Colorado** is largest; world's largest plateau remains **Tibetan**.

N.2A Valleys

A valley is a long lowland between higher ground. Rivers, ice, rifts and solution each leave a different valley signature.

Valley types and named valleys — Teaching Card

How valleys form

- A **rift valley** forms where the crust is pulled apart along parallel faults and a central block drops — long, steep-sided, and structurally controlled.
- A **blind valley** ends abruptly where a stream sinks underground — a **karst** signature with sinkholes (**dolines**) nearby.
- Most ordinary valleys are river-cut; glacial troughs are broader **U-shaped** cuts (taught with glacial landforms).

Named valleys and basins (match-list centre)

- **Death Valley** sits mainly in **California**, USA. It is a **rift / block** valley in the Basin-and-Range province, holds the Devil's Golf Course salt pan, includes the lowest point in North America (about **−86 m**), and is famous for extreme **heat** — not a cold Siberian lowland.
- The **Panjshir Valley** lies in north-eastern **Afghanistan**, in the Hindu Kush north of Kabul.
- **Taylor Valley** is a dry valley in **Antarctica** — the coaching tag "Valley of the Dead" follows Scott's 1903 note.
- **Silicon Valley** is the technology / chip belt south of San Francisco Bay in **California** — an economic name, not a rift landform.
- The **Valley of the Kings** is the pharaoh burial ground on the west bank of the Nile near Luxor in **Egypt** (मिस्र).
- The **Great Artesian Basin** is a huge groundwater basin in **Australia** (about **1.7 million km²**) between the Great Dividing Range and the interior uplands — a basin of stored water, not a single river gorge.
- The **Great Basin** of the **USA** is **endorheic** (internal drainage) between the Rockies and the Sierra Nevada, with the Columbia Plateau to the north and Mojave / Sonoran deserts to the south.
- **Blind valleys** and **sinkholes (dolines)** belong to **karst** limestone landscapes, not to ordinary river plains.

Place	Quick recall
Death Valley	California rift; heat; salt pan; ~−86 m
Panjshir	NE Afghanistan (Hindu Kush)
Taylor Valley	Antarctica — "Valley of the Dead"
Silicon Valley	California chip / tech belt
Valley of the Kings	Egypt (Luxor / Nile west bank)
Great Artesian Basin	Australia groundwater basin
Great Basin	USA endorheic interior
Blind valley / doline	Karst limestone

N.3 Plains

Plains are **low, gently rolling** lands — the world's grain and cattle belts.

How plains form — Teaching Card

- An **alluvial** (जलोढ़) plain is built by **river dump** of silt and sand — Indo-Gangetic, Nile, Mesopotamia (मेसोपोटामिया), Hwang Ho and Mississippi.
- A **glacial / till** plain is left by an ice sheet's debris — North European Plain and the Prairie of Canada–USA.
- A **coastal** plain mixes **sea and river** work — Atlantic coastal USA and eastern India.
- A **lacustrine** plain is an **old lake floor** — Imphal-type basins and some interior lows.
- Pampas vs Campos share an "open land" idea but sit in **different countries** — **Argentina** vs **Brazil**.

Kind	How formed	Example
Alluvial	River dump	Indo-Gangetic, Nile, Mesopotamia, Hwang Ho, Mississippi
Glacial / till	Ice-sheet	North European Plain, Prairie of Canada–USA
Coastal	Sea + river	Atlantic coastal USA; eastern India
Lacustrine	Old lake floor	Imphal-type locally; some interior basins
Plain	Where	Note
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Indo-Gangetic	India–Pakistan–Bangladesh	Alluvium; wheat–rice–cane
North / Great European	France → Poland → Russia	Wheat–dairy; glacial + river
Great Plains / Prairie	USA–Canada	Temperate grassland overlay

Kind	How formed	Example
Pampas	**Argentina** (into Uruguay)	Temperate grassland **and** a plain. **Not Chile, not Brazil**
Llanos	Orinoco (Venezuela–Colombia)	Tropical grassland plain
Manchurian	NE China	Soy / wheat
West Siberian	Russia	Ob–Yenisei lowland
Mesopotamian	Iraq	Tigris–Euphrates
Nile valley / delta (डेल्टा)	Egypt	Alluvium in a desert
Po valley	N Italy	Alpine alluvium
Hungarian (Alföld)	Hungary	**Puszta** grassland sits here
Canterbury	New Zealand	Downs / sheep
Nullarbor	S Australia	Limestone plain, not a grassland name

N.4 Volcanoes (world locations)

Four belts (location, not process)

Belt	Share of great volcanoes	names
Circum-Pacific / Ring of Fire	**~80%**	Fuji, Pinatubo, Mayon, Apo, St Helens, Rainier, Popocatépetl, Andes cones, Krakatoa, Tambora
Mid-Atlantic	Divergent	**Iceland** (Hekla), Azores
Alpine–Himalayan / Med	Collision	**Etna, Vesuvius, Stromboli** (Italy)
East African Rift	Tension	**Kilimanjaro, Kenya — not** Ring of Fire
Volcano	Country	Belt / extra
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Fuji	Japan	Pacific. Classic cone
Pinatubo	Philippines	Pacific. 1991 eruption tag
Mayon	Philippines	Pacific. Perfect cone
Apo	Philippines	Highest PH; 2021 list
St Helens	USA (Washington)	Cascades / Pacific. 1980

Belt	Share of great volcanoes	names
Rainier	USA	Cascades / Pacific. 2021 list
Popocatépetl	Mexico	Pacific
Paricutin (paper: **Pericutine**)	**Mexico**	Cinder cone; born 1943. 2021 list
Cotopaxi	Ecuador	Andes / Pacific rim
Krakatoa / Krakatau	Indonesia	Sunda arc / Pacific
Tambora	Indonesia	1815; “year without summer”
Mauna Loa / Mauna Kea	Hawaii (USA)	**Hot-spot shield**, not a collision arc
Etna	**Italy** (Sicily)	Med. 2021 list. **Not** Pacific
Vesuvius	Italy	Pompeii AD 79
Stromboli	Italy	“Lighthouse of the Mediterranean”
Kilimanjaro	Tanzania	**Rift**. 2024 “which is NOT Ring of Fire”
Mt Kenya	Kenya	Rift. Lower than Kilimanjaro
Erebus	Antarctica	Ross Island; southern active
Hekla	Iceland	Mid-Atlantic

> **Logic:** ** For **Pacific Ring of Fire** lists, exclude **Kilimanjaro** (and Etna if it appears). Paper spelling **Pericutine** = Paricutin, Mexico.

Inline PYQ — UPPCS Prelims 2024, Q30

Which of the following volcanoes is not located in the Pacific Ring of Fire?

Options:

A. Mount Fuji

B. Mount Pinatubo

C. Mount Kilimanjaro

D. Mount St. Helens

Show answer

Ans: C — Kilimanjaro sits on the **East African Rift**. A, B, D are Pacific-rim.

Inline PYQ — UPPCS Prelims 2021, Q78

Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Volcanic Mountains)	List-II (Country)
A. Mount Rainier	1. Italy
B. Mount Etna	2. Mexico
C. Mount Pericutine	3. Philippines
D. Mount Apo	4. U.S.A.

Options:

A. 4 2 1 3

B. 4 1 2 3

C. 2 1 4 3

D. 4 3 2 1

Show answer

Ans: B — A-4 USA · B-1 Italy · C-2 Mexico · D-3 Philippines. Trap **A** gives Etna to Mexico (that is Paricutin). Trap **D** gives Etna to the Philippines (that is Apo).

N.5 Islands

Australia is a continent, not an island. Rank **islands** by area (standard coaching / atlas order; Australia stays off this list):

Rank	Island	Note
1	**Greenland**	Largest island; Denmark politically
2	**New Guinea**	Indonesia (Papua) + Papua New Guinea
3	**Borneo**	Indonesia, Malaysia, Brunei
4	**Madagascar**	Africa’s large island
5	**Baffin**	Canada (Nunavut)
6	**Sumatra**	Indonesia
7	**Honshu**	Japan’s main island
8	**Victoria**	Canada (Arctic)
9	**Great Britain**	England–Scotland–Wales
10	**Ellesmere**	Canada (Arctic)

Chain: Greenland → New Guinea → Borneo → Madagascar → Baffin → Sumatra → Honshu → Victoria → Great Britain → Ellesmere.

Kind	How made	Examples
Continental	Piece of a continent, shallow shelf	**Greenland**, **Borneo**, **New Guinea**, **Madagascar**, Britain, Sri Lanka
Volcanic arc / hot-spot	Magma	**Java**, **Iceland**, **Hawaii**, Japan (Honshu mixed arc), Sicily (Etna)
Coral / atoll	Reef on a sinking volcanic peak or shelf	Maldives, Lakshadweep (लक्षद्वीप), many Pacific atolls
Coral reef (not an island rank)	—	Great Barrier Reef (Australia) is a **reef**, not a "largest island" option
Island	Fact	
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Greenland	**Largest island**. Denmark. Ice sheet. Inuit	
New Guinea	2nd. Split **Indonesia (Papua) + Papua New Guinea**	
Borneo (Kalimantan)	3rd. **Three countries: Indonesia, Malaysia (Sabah/Sarawak), Brunei**. **Sunda shelf — not** volcanic deposition (2022 **Only 1**)	
Madagascar	4th. Africa's big island; unique fauna (प्राणीजात); **not** three-country split	
Baffin	Canada / Arctic	
Sumatra	Indonesia; Sunda arc **is** volcanic — contrast with Borneo	
Java	Indonesia; dense, volcanic	
Sulawesi	Indonesia; K-shaped	
Honshu	Japan's main island. Hokkaido / Kyushu / Shikoku = the other three	
Victoria	Canada / Arctic; after Honshu in the area ladder	
Great Britain	England–Scotland–Wales (Ireland is separate)	
Ellesmere	Canada / Arctic; after Great Britain in the top-ten set	
Iceland	Mid-Atlantic Ridge; volcanic	
Hawaii	Pacific hot-spot chain	

Kind	How made	Examples
Sri Lanka	Continental; Palk Strait / Gulf of Mannar vs India	
Tasmania	S of Australian mainland	
Sicily / Sardinia / Corsica / Crete / Cyprus	Med islands. Sicily has **Etna**	
New Zealand	North Island + South Island; Cook Strait	

Greater Sunda: Sumatra, Java, **Borneo**, Sulawesi. Only Borneo is the "three countries, not volcanic dump" PYQ.

> **Logic:** ** Stmt 1 true, stmt 2 false for Borneo. Java/Sumatra **are** volcanic-arc; do not copy that onto Borneo.

World island desk (GC extras)

Island / group	Note
Nuuk	Capital of Greenland; elevation ~**1–3 m**
Diego Garcia	Largest of **Chagos** (Indian Ocean); UK/US base
Socotra	**Yemen**; UNESCO natural heritage **2008**; Guardafui / Arabian Sea mouth — **not Oman**
Seychelles / Mauritius / Madagascar	SW Indian Ocean set
Falklands	**South Atlantic** (UK), ~650 km E of S America — **not** Indian Ocean
Azores / Madeira / Cape Verde	Atlantic (Azores/Madeira = Portugal; Cape Verde = sovereign)
Bahamas	Atlantic / Lucayan
Fiji	**Pacific** Melanesia; Viti Levu + Vanua Levu hold most people
Melanesia	New Guinea, Bismarck, Solomon, Vanuatu, New Caledonia, **Fiji** (not Gilbert/Marshall (मार्शल)/Society = Micronesia/Polynesia)
Galápagos	Pacific volcanic; **Ecuador** (not Peru)
Grenada	Caribbean; "Island of Spices"
Balearic	Spain; **W Mediterranean**; Palma de Mallorca
Faroe	Denmark; **Sheep Islands**; N Atlantic
Macau	Handed to China **1999**
Majuli	Largest river island; **Brahmaputra** (ब्रह्मपुत्र), Assam

Island / group	Note
Japan N→S	**Hokkaido** → **Honshu** → **Shikoku** → **Kyushu**; Tokyo on **Honshu** (largest)
Indonesia W→E	**Sumatra** → **Java** → **Bali (बलि)** → **Lombok**
NZ South Island	Larger than NZ North / Java / Newfoundland among common options
Australia	**Island continent** (smallest continent)
Polynesia	Tonga, Tuvalu, Tahiti — **Tinian** (Marianas) is **not** Polynesia
Saint Martin	Caribbean salt-pan / lagoon coast pair
Indonesia island count	~**17,500** islands (highest among common MCQ options; Sweden note for absolute (निरपेक्ष) world count)

Inline PYQ — UPPCS Prelims 2022, Q126

With reference to Borneo Island, located in South-East Asia, which of the following statements is/are true?

1. It is divided among three countries.
2. It has been formed due to volcanic deposition.

Options:

A. Neither 1 nor 2

B. Both 1 and 2

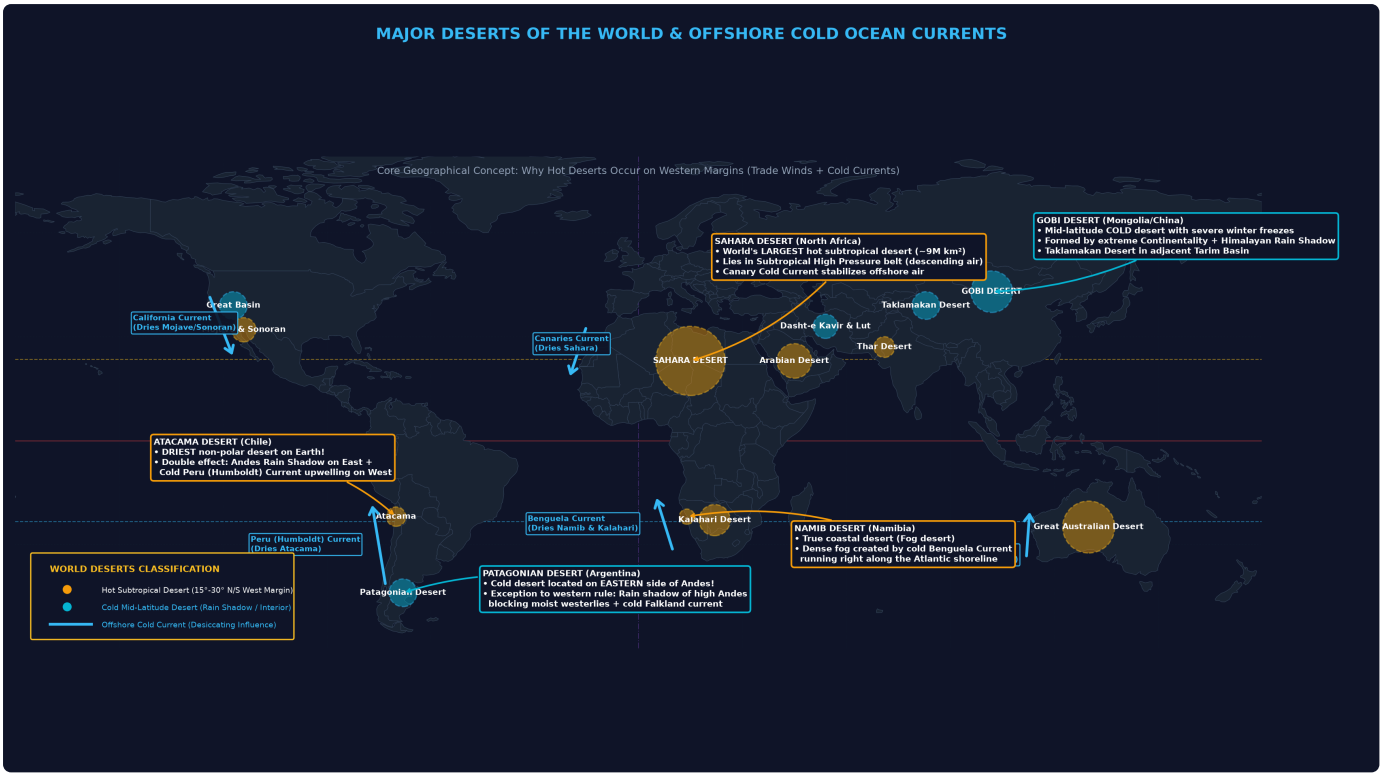
C. Only 2

D. Only 1

Show answer

Ans: D — Indonesia + Malaysia + Brunei. It sits on the **Sunda continental shelf**, not a Hawaii-type volcanic pile. Trap **B** is the “all Indonesia = volcano” dump.

N.6 Deserts



Deserts are arid lands (often **< 25 cm** rain) with sparse xerophytic vegetation.

Why hot deserts favour west coasts

- Subtropical **Horse Latitudes** (descending dry air of the subtropical high).
- **Cold ocean currents** on western margins suppress convection (fog possible, rain rare).
- Classic set: **Atacama** (Humboldt / Peru), **Namib** (Benguela), Sahara–Arabian west-margin logic, plus Thar as a monsoon-margin / rain-shadow cousin.

Kind	Why dry	Examples
Hot subtropical	Descending air ~**30°** (horse latitudes)	Sahara, Arabian, Australian, Kalahari, Thar
Coastal / fog	Cold current + stable air	**Atacama** (Peru/Humboldt), **Namib** (Benguela)
Rain-shadow	High range steals moisture	**Patagonia** (Andes), **Atacama** (Andes), Thar (Aravalli leeward)
Cold interior	Far from sea, winter freeze	**Gobi**, **Taklamakan**, Great Basin, Colorado Plateau dry side
Desert	Country / belt	
--------	----------	------
Sahara	N Africa (many states)	**Largest hot** desert. Hoggar / Tibesti are highland islands in it
Arabian	Arabian Peninsula	2nd large hot desert. **Rub' al-Khali** = Empty Quarter

Kind	Why dry	Examples
Gobi	**Mongolia + China** (Inner Mongolia, Gansu, Xinjiang fringe)	**Cold**. 2025: **Only 1 and 2**. **Not** Russia, **not** Kazakhstan
Great Sandy	**NW Australia**	Smallest of the 2023 four
Great Victoria	S/W Australia	Largest **Australian** desert by many lists
Gibson	**Australia** (between Victoria and Sandy)	**Not Brazil** (2018)
Simpson	Central Australia	Red dunes
Sonoran	**USA** + Mexico	2018 correct pair. Cactus desert
Mojave	USA	Death Valley
Chihuahuan	USA–Mexico	
Taklamakan	**China** (Xinjiang)	Cold; Tarim basin. 2018 correct pair
Karakum	**Turkmenistan**	2018 correct pair. **Not** Kazakhstan's default dump
Kyzylkum	Uzbekistan–Kazakhstan	Red sand; separate from Karakum
Kalahari	Botswana / South Africa / Namibia	Bushman / San. Semi-desert
Namib	Namibia coast	Fog; among the oldest
Atacama	N **Chile** (into Peru)	Among the **driest** on Earth
Patagonia	**Argentina**	Cold rain-shadow. Not Chile's Atacama
Thar	India–Pakistan	Pleistocene–recent sands
Dasht-e-Lut / Kavir	Iran	Hot interior; Lut = extreme surface-temperature renown (NASA Aqua note)
Great Basin	USA	Cold interior
Rubʿ al-Khali	Arabian Peninsula	Largest contiguous sand sea
An Nafud	Saudi Arabia	

2023 size ladder (smallest → largest): Great Sandy → **Gobi** → **Arabian** → **Sahara**. Code **I, IV, II, III**.

Extra GC pairs - Deserts usually receive **under 25 cm** of annual rain. - Hot deserts sit on **western** continental margins near **15–30°**. - Knies note — deserts receive more solar energy in about 6 hours than humanity uses in a year. - The **Thar** is the most densely populated desert. - **Europe** has the least desertification problem among continents (essentially no true hot desert). - The Kalahari sits in the **Southern Hemisphere** (Botswana, Namibia and South Africa). - The African–Eurasian desert belt sits under subtropical highs — **not** mainly because of warm currents.

> **Logic:** ** Kazakhstan has **other** dry belts (Kyzylkum fringe, steppe). That does **not** put Gobi into option 4.

Inline PYQ — UPPCS Prelims 2025, Q27

The Gobi Desert extends into which of the following countries?

1. Mongolia
2. China
3. Russia
4. Kazakhstan

Select the correct answer from the code given below:

Options:

A. Only 1 and 2

B. Only 1, 2 and 4

C. Only 1, 2 and 3

D. Only 1

Show answer

Ans: A — Core Gobi = **Mongolia + northern China**. Trap **B** adds Kazakhstan; trap **C** adds Russia; trap **D** drops China.

Inline PYQ — UPPCS Prelims 2023, Q63

Consider the following deserts and arrange them in increasing order of their size (from smallest to largest).

(I) Great Sandy Desert

(II) Arabian Desert

(III) Sahara Desert

(IV) Gobi Desert

Options:

A. I, II, IV, III

B. II, I, III, IV

C. I, IV, II, III

D. II, I, IV, III

Show answer

Ans: C — Great Sandy < Gobi < Arabian < Sahara. Trap **A** puts Arabian before Gobi. Trap **B** starts with Arabian (too big).

Inline PYQ — UPPCS Prelims 2018, Q27

Which of the following pairs is NOT correctly matched?

(Desert — Country)

Options:

A. Sonoran – USA

B. Taklamakan – China

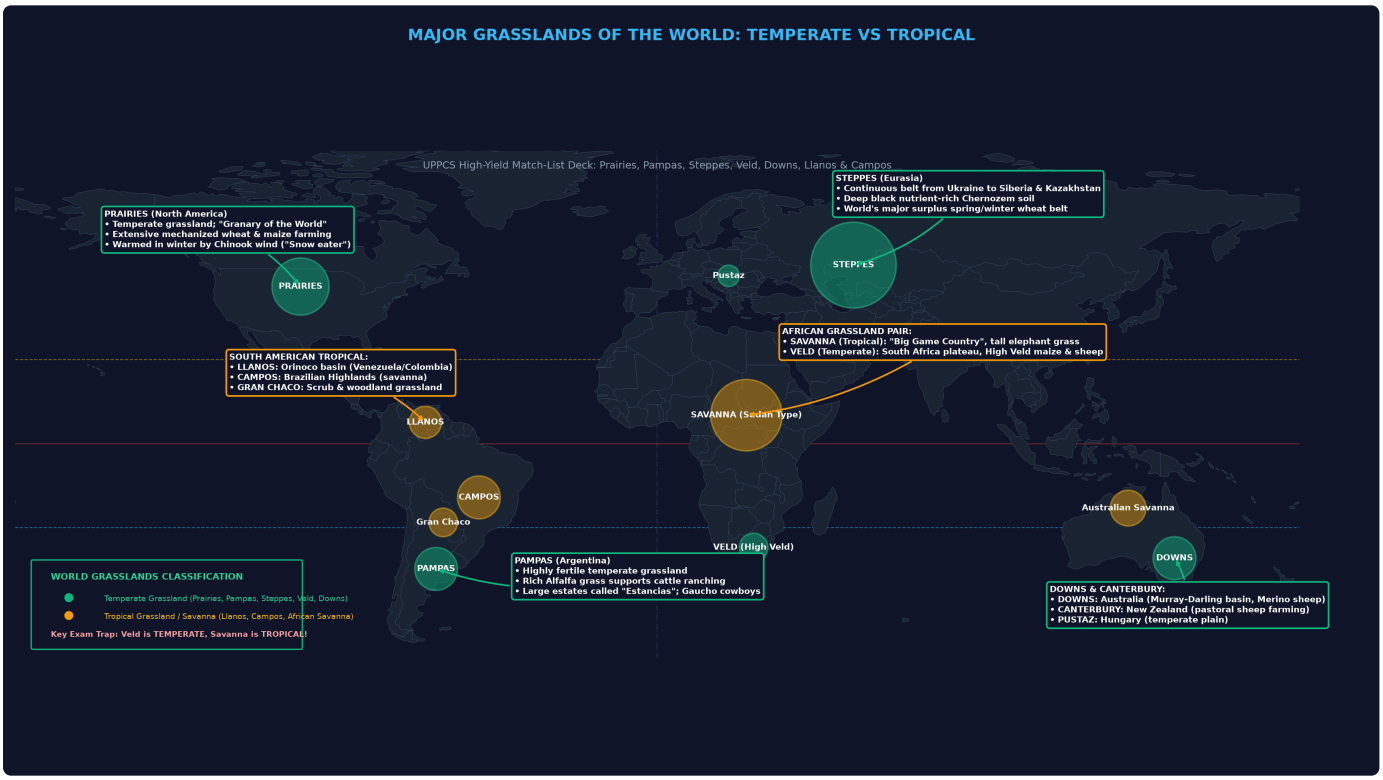
C. Karakum – Turkmenistan

D. Gibson – Brazil

Show answer

Ans: D — **Gibson** is in **Australia**. A, B, C are standard correct pairs. Brazil's dry name s is more often **Caatinga** (scrub), not Gibson.

N.7 Grasslands



Tropical grassland is **savanna** — summer rain with scattered trees.

Temperate grassland covers prairie, pampas, steppe, veld and downs — hot summers, cold winters and few trees.

- Temperate grasslands often sit on fertile **chernozem / black earth** (especially Eurasian **steppe** and North American **prairie**) — the world's wheat belts.
- They are treeless mainly because rainfall is **too low for forest**, not because it is always freezing.

Local name	Where	Climate	Economy tag
Prairie	USA–Canada (Great Plains)	Temperate	Wheat / maize / cattle; chernozem-type soils
Pampas	**Argentina** (+ Uruguay)	Temperate	Cattle / wheat
Veld (High / Middle / Low)	**South Africa**	Temperate–subtropical	Maize, gold belt overlay
Downs	**Australia**	Temperate	Sheep / wheat
Canterbury Downs	New Zealand	Temperate	Sheep
Steppe	Ukraine–S Russia–Kazakhstan–Mongolia	Semi-arid **BS**	Wheat / nomadism; classic **chernozem**
Puszta	**Hungary**	Temperate	Hungarian Alföld fact
Campos	**Brazil** (plateau)	Tropical / temperate mix	Cattle. **Not** Pampas
Llanos	**Venezuela–Colombia** (Orinoco)	Tropical	Cattle
Savanna / Sudan type	Africa (also N Australia, Brazilian edges)	Tropical **Aw**	**Land of Big Games**
Sahel	S of Sahara	Semi-arid fringe	Not the same as savanna core
Gran Chaco	Paraguay–N Argentina–Bolivia	Dry woodland/grass	Not Pampas

The label **“Land of Big Games”** belongs to the **Tropical Savanna**. Temperate grassland is farms and cattle, not the lion (सिंह)–zebra poster. Hot desert is **not** this label.

Latitude pairs: Tropical savanna sits near **10–25°** N/S (between the tropics). Temperate grasslands sit near **30–45°** (and can stretch toward **50°N** where the landmass is wide).

Selvas (Selva) names the Amazon **tropical rainforest**, not a grassland. **Downs** are **temperate** Australian grasslands, not tropical.

Savanna forest check (IAS-type): trees stay sparse mainly because of **fire, grazing herbivores and seasonal rainfall** (not burrowing animals or soil alone as the lead trio).

Pampas A/R: fertile soils plus temperate climate make Argentina’s main cultivation and cattle region — Reason correctly explains Assertion when both are true.

Paper spelling **Welds / Velds** means South Africa. **Pustaz / Puszta** means Hungary.

Inline PYQ — UPPCS Prelims 2018, Q26

In which of the following countries is the Pampas grasslands situated?

Options:

A. Argentina

B. Brazil

C. Chile

D. Ecuador

Show answer

Ans: A — Brazil = **Campos**. Chile = **matorral** / Andes rain-shadow (Atacama–Patagonia), not Pampas. Ecuador = Andes / Chimborazo, not a pampas country.

Inline PYQ — UPPCS Prelims 2019, Q10

Which one of the following natural regions is known as "Land of Big Games"?

Options:

A. Temperate Grassland

B. Tropical Monsoon Region

C. Hot Desert Region

D. Tropical Savanna Region

Show answer

Ans: D — African savanna (Serengeti-type). Trap **A** is prairie/pampas. Trap **C** is Sahara safari (वन्यजीव अभयारण्य) confusion.

N.8 Vegetation (world belts)

Forest cover (वन आवरण) pairs (Bhutan forest constitution) - Bhutan's constitution targets ≥**60%** forest (actual cover is higher). - Among common options Japan has a high percentage cover. - The softwood/conifer belt is the **taiga** (largest forest-area type). - Amazon **Selva** is broadleaf evergreen rainforest. - Tropical rainforest soil is **nutrient-poor** despite high productivity. Cleared rainforest does **not** stay fertile like temperate clearings.

Walk **equator** → **pole** (one row each):

Belt	Where	Plant picture
Equatorial rainforest (Af)	Amazon, Congo, Malaysia–Indonesia	Multilayer, **epiphytes**, hardwood (mahogany, ebony). Brazil name = **Selva**
Tropical monsoon	India, mainland SE Asia, N Australia	Teak, sal; sheds in dry season
Savanna	Sudan belt, Campos, Llanos, N Australia	Grass + scattered trees; **baobab**
Hot desert / thorn	Sahara, Thar, Australia	**Acacia**, cactus, xerophytes

Belt	Where	Plant picture
Mediterranean sclerophyll	5 west-coast patches (N.9)	Olive, vine, cork, **cedar**; shrubs below
China type / laurel	E coasts ~25–40°	Camphor, tea in hills; summer rain forest
Temperate deciduous	W Europe, E USA, N China	Oak, beech, maple
Taiga / boreal	Canada, Scandinavia, Siberia	**Conifer** — spruce, fir, pine, larch. **Not** treeless
Tundra	Arctic fringe, high Andes/Tibet alpine	**Treeless** — moss, lichen, dwarf shrub. **Cryophyte steppe** = Arctic tundra (2024)
Ice	Greenland, Antarctica	None

Same climate, four local shrub names:

Name	Region
Maquis	**Mediterranean** Sea coast
Fynbos (paper: **Fynbus**)	**South Africa** (Cape)
Chaparral	**California**
Matorral	**Chile**

Code **2 3 1 4**.

Plant → region:

Plant	Region
Epiphytes	Equatorial
Acacia	Sahara / thorn
Baobab	**Savanna**
Cedars	**Mediterranean**

Code **2 3 4 1**.

Brazil pair	Meaning
Selva	Equatorial **rainforest**
Terra Roxa	Fertile dark **basaltic coffee soil** — **not** a forest name. Parallel to Deccan fertility, **not** the same as Indian regur (रेगुर) as a name

Other tags

- **Planting hearth** is C.O. **Sauer** (origin of agriculture).

- **World floral regions** is Adolf **Engler**.
- **Puszta** is Hungary (grassland, not a forest).
-

Inline PYQ — UPPCS Prelims 2019, Q1

Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Vegetation Type)	List-II (Region)
A. Maquis	1. California
B. Fynbus	2. Mediterranean Sea Coastal Region
C. Chaparral	3. South Africa
D. Matorral	4. Chile

Options:

A. 2 3 1 4

B. 4 3 2 1

C. 3 4 1 2

D. 4 2 3 1

Show answer

Ans: A — Maquis–Med · Fynbos–South Africa · Chaparral–California · Matorral–Chile. Trap **B** starts Maquis on Chile. Trap **D** gives Fynbos to the Mediterranean.

Inline PYQ — UPPCS Prelims 2019, Q3

Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Natural Vegetation)	List-II (Region)
A. Epiphytes	1. Mediterranean
B. Acacia	2. Equatorial
C. Buobab	3. Sahara
D. Cedars	4. Savanna

Options:

A. 2 3 4 1

B. 2 3 1 4

C. 2 4 3 1

D. 2 4 1 3

Show answer

Ans: A — Epiphytes–Equatorial · Acacia–Sahara · Baobab–Savanna · Cedars–Mediterranean. Trap **C** swaps baobab onto Sahara (baobab is savanna). The paper spelling **Buobab** means baobab.

Inline PYQ — UPPCS Prelims 2025, Q63

With reference to Brazil, which of the following statements is/are correct?

1. The equatorial rainforests of Brazil are called "Selva".
2. The fertile dark-coloured soil found there is known as "Terra Roxa".

Select the correct answer from the code given below:

Options:

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: C — Both are Brazil facts. Selva is the **forest**; Terra Roxa is the **soil**.

Inline PYQ — UPPCS Prelims 2024, Q29

Match List-I with List-II and choose the correct answer using the codes given below the lists:

List-I	List-II
A. Planting Hearth	1. Hungary
B. Cryophyte Steppe	2. Adolf Engler
C. World Floral Regions	3. C.O. Sauer
D. Puszta Grassland	4. Arctic Tundra

Options:

A. 3 4 2 1

B. 2 1 4 3

C. 3 4 1 2

D. 2 3 4 1

Show answer

Ans: A — Sauer · Arctic tundra · Engler · Hungary. Trap **C** swaps Engler and Hungary.

N.9 World Climatic Regions

Pressure belts **follow the Sun**. At the **winter solstice** (NH December) they shift **south** — that is why Mediterranean west coasts get **winter** westerlies.

Type (Köppen (कोपेन))	Where	Rain timing	Vegetation / crop
Equatorial (Af)	Amazon, Congo, Malaysia	**Year-round** convection	Selva; **not** a citrus belt
Savanna / Sudan (Aw)	Africa belt, Campos, Llanos, N Australia	**Summer** wet, winter dry	Grass + trees; baobab; **Land of Big Games**
Tropical monsoon	India, mainland SE Asia	**Summer** monsoon	Deciduous teak/sal
Hot desert (BWh)	Sahara, Arabia, Thar, Australia	Rare	Xerophytes
Cold desert (BWk)	Gobi, Taklamakan	Rare / snow	Sparse
Steppe (BS)	Ukraine to Mongolia; rain-shadow edges	Scanty	Short grass
Mediterranean (Cs)	**30–45° west coasts** — five patches below	**Winter** cyclonic	Olive, citrus, vine, cork, wheat, cedar, maquis
China / warm-temperate east coast	E China, SE USA, NSW, Natal	**Summer** rain	Laurel / tea hills
British / W European (Cfb)	Britain, NW France, W Norway	**Year-round** cyclonic	Deciduous oak–beech
Taiga / boreal	50–70°N	Modest, summer max	Conifers
Tundra (ET)	Arctic coast	Low	Moss/lichen; permafrost (स्थायी हिम मृदा)
Ice cap (EF)	Greenland interior, Antarctica	—	None

Five Mediterranean patches (all **west** coasts, not east):

1. Mediterranean Sea basin (Spain, Italy, Greece, Levant (लेवेंट), Maghreb)
2. **California**
3. Central **Chile**
4. SW **Australia** (Perth side)
5. **Cape** / Western Cape, South Africa

Mediterranean mechanism

- **Summer:** subtropical **high** + **dry terrestrial winds** → drought (सूखा) (bright, good for citrus/vine).

- **Winter:** belts shift equatorward → **westerlies** + Mediterranean cyclones → rain.
- Do not give Mediterranean **summer** rain (that is monsoon / China type).

Inline PYQ — UPPCS Prelims 2024, Q79

Consider the following statements with reference to Mediterranean climate:

1. It receives rainfall in winter season.
2. Due to winter solstice, air pressure belts shift to the south.

Which of the above statements is/are correct?

Options:

A. Only 2

B. Neither 1 nor 2

C. Only 1

D. Both 1 and 2

Show answer

Ans: D — Winter rain is the definition. Belts follow the Sun, so NH winter solstice → shift **south** → westerlies over the Med. Trap **C** drops the belt-shift, which is the *why*.

Inline PYQ — UPPCS Prelims 2022, Q144

Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Rainfall occurs in the winter season in Mediterranean climate.

Reason (R): In summer these regions remain under the influence of dry terrestrial winds.

Options:

A. (A) is false but (R) is true.

B. (A) is true but (R) is false.

C. Both (A) and (R) are true but (R) is not the correct explanation of (A).

D. Both (A) and (R) are true and (R) is the correct explanation of (A).

Show answer

Ans: D — Summer dry high / dry land winds keep the rain for winter. That is why the climate is “winter-rain”.

Inline PYQ — UPPCS Prelims 2020, Q72

In which of the following regions of the world, the production of citrus fruits is well developed?

Options:

A. Between Kangra (कांगड़ा) and Dhauladhar (धौलाधार) ranges

B. Between Pir (पीर) Panjal (पीर पंजाल) and Himadri (हिमाद्रि) ranges

C. Mediterranean regions

D. Equatorial regions

Show answer

Ans: C — Winter rain + bright dry summer. Equatorial is too wet and cloudy. A/B are Himalayan valley distractors, not the world citrus belt.
